GLOBAL SUPERSTORE
BUSINESS PERFORMANCE ANALYSIS

Professional Integrated SQL & Power BI Business Report

Analysis Period: 2012–2015

Executive KPI	Result
Total Sales	$12.64M
Total Profit	$1.47M
Profit Margin	11.61%
Total Orders	25,728
Products Sold	178,312
Unique Customers	17,415
Unique Products	3,788

Executive Summary

Global Superstore generated $12.64 million in sales and $1.47 million in profit during 2012–2015, producing an overall profit margin of 11.61%. The business processed 25,728 orders and sold 178,312 products to 17,415 unique customers across 3,788 products.

The integrated analysis combines PostgreSQL SQL findings with two Microsoft Power BI dashboards. Technology is the strongest category, while Furniture generates substantial sales but only a 6.94% margin. Western Europe is the strongest major region, while Canada achieved a 26.61% regional margin in the SQL analysis.

The key management message is that sales growth alone is not sufficient. Product margin, discounts, shipping costs, regional performance, customer profitability and seasonal demand should be managed together to improve profitable growth.

1. Project Overview, Business Problem and Methodology

1.1 Objective

The objective was to convert Global Superstore transactional data into practical business intelligence that supports management decisions concerning sales, profitability, customers, products, regions and operational efficiency.

1.2 Business Questions

- What are total sales, profit, margin and order activity?
- Which months produce the strongest sales and profit?
- Which categories and products create the most value?
- Which products generate losses?

- Which regions and customers are most valuable?
- How should discounts and shipping costs be controlled?
- Where are the strongest opportunities for profitable growth?

1.3 Dataset and Tools

Item	Detail
Source	Kaggle
Rows	51,290
Columns	24
Period	2012–2015
Preparation	Microsoft Excel
Analysis	PostgreSQL
Visualization	Microsoft Power BI

1.4 Analytical Workflow

1. Acquire and review the Kaggle dataset.
2. Prepare and validate the data in Excel.
3. Load and analyze the prepared data in PostgreSQL.
4. Create analytical views for executive, monthly, category, product, regional and customer analysis.
5. Build Power BI dashboards from the analytical results.
6. Interpret findings and translate them into recommendations.

1.5 SQL Techniques

The analysis used SUM(), COUNT(), COUNT(DISTINCT), AVG(), GROUP BY, ORDER BY, HAVING, LIMIT, DATE_TRUNC(), NULLIF(), aggregate calculations and profit-margin calculations.

1.6 Project Deliverables

- PostgreSQL SQL analysis and analytical views
- Microsoft Power BI Executive Dashboard
- Microsoft Power BI Profitability & Performance Dashboard
- Integrated professional business report
- Microsoft Excel data preparation workflow

2. Executive Performance and Time Analysis

The Executive Dashboard provides a management-level view of sales, profit, margin, orders, monthly trends, leading regions and category performance.

Executive Dashboard KPI	Value
Total Sales	$12.64M
Total Profit	$1.47M
Profit Margin	11.61%
Total Orders	25.7K

The business is profitable overall, but the 11.61% margin demonstrates why management should monitor the quality and cost of sales rather than focusing only on revenue growth.

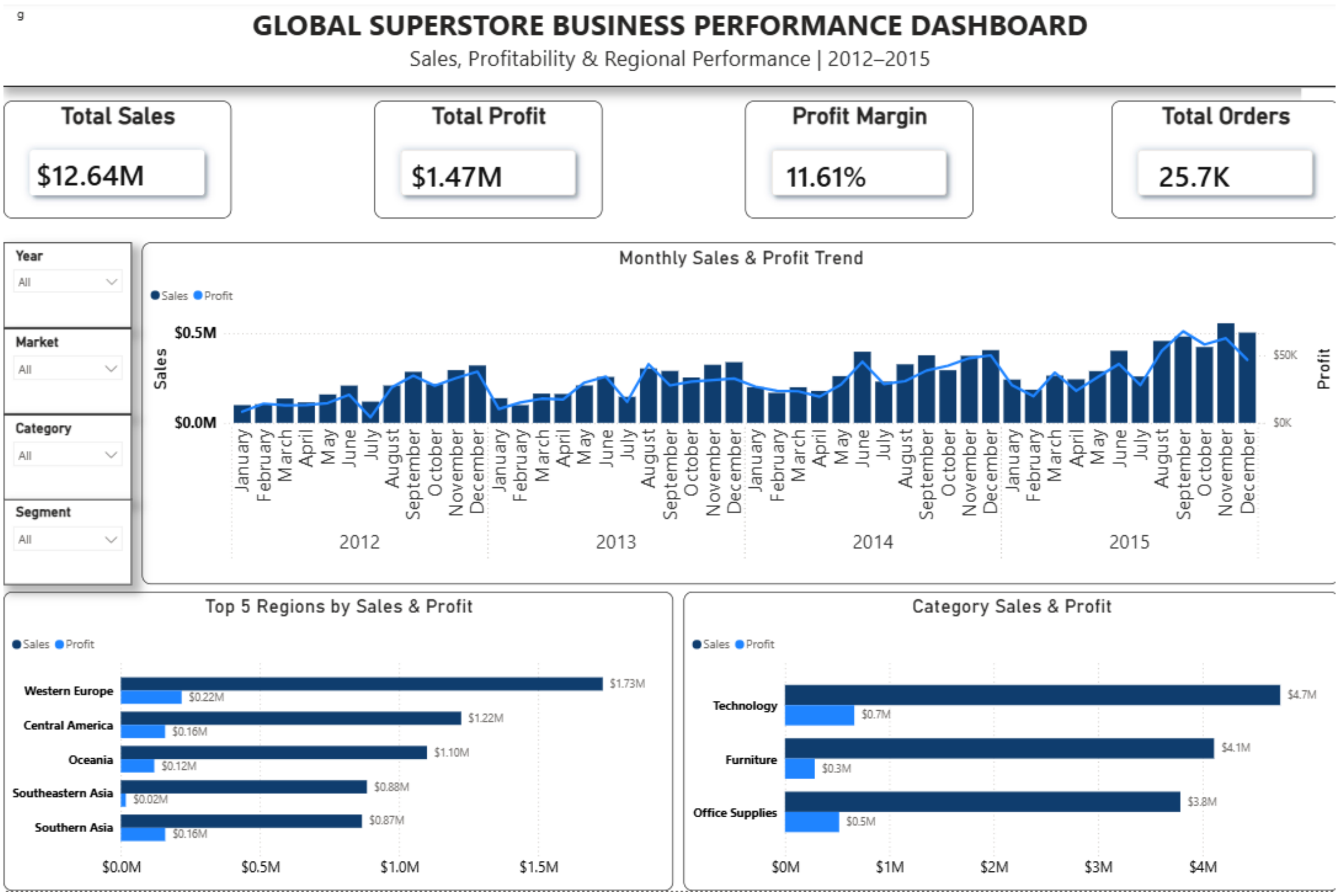

Figure 1: Global Superstore Executive Business Performance Dashboard

2.1 Monthly Performance

Metric	Best Period	Result
Highest Sales	November 2015	$555,312
Highest Profit	September 2015	~$68,000
Highest Orders	December 2015	2,153
Highest Quantity	November 2015	7,706
Highest Shipping Cost	November 2015	$60,160

November 2015 generated the highest sales and quantity sold, while September 2015 generated the highest monthly profit. December 2015 recorded the highest number of orders. The different peak months show that higher sales volume does not automatically result in higher profit.

2.2 Management Interpretation

- Investigate what made September 2015 unusually profitable.
- Prepare inventory and logistics capacity for high-demand periods.
- Evaluate product mix and discounting during peak months.
- Use monthly profit and margin alongside sales when evaluating performance.

3. Category and Product Profitability

3.1 Category Performance

Category	Sales	Profit	Margin	Quantity	Avg. Discount
Technology	$4.74M	$663.8K	13.99%	35,176	11.20%
Office Supplies	$4.11M	$518.7K	13.69%	108,244	14.71%
Furniture	$3.79M	$285.2K	6.94%	34,892	11.34%

Technology is the strongest category across sales, profit and margin. Office Supplies has the highest quantity sold and highest average discount, making discount effectiveness an important management question. Furniture is the main category-level profitability concern because it produces almost $3.8 million in sales but only a 6.94% margin.

3.2 Profitability & Performance Dashboard

KPI	Value
Total Sales	$12.64M
Total Profit	$1.47M
Profit Margin	11.61%
Average Discount	13.37%

The Profitability & Performance Dashboard extends the executive analysis by examining category and product profitability, discounting and regional shipping costs. The dashboard includes an Average Discount KPI, a Discount vs. Profitability visual, and a Top 5 Shipping Cost by Region visual. These additions support deeper investigation of factors that may affect profitability and operational efficiency.

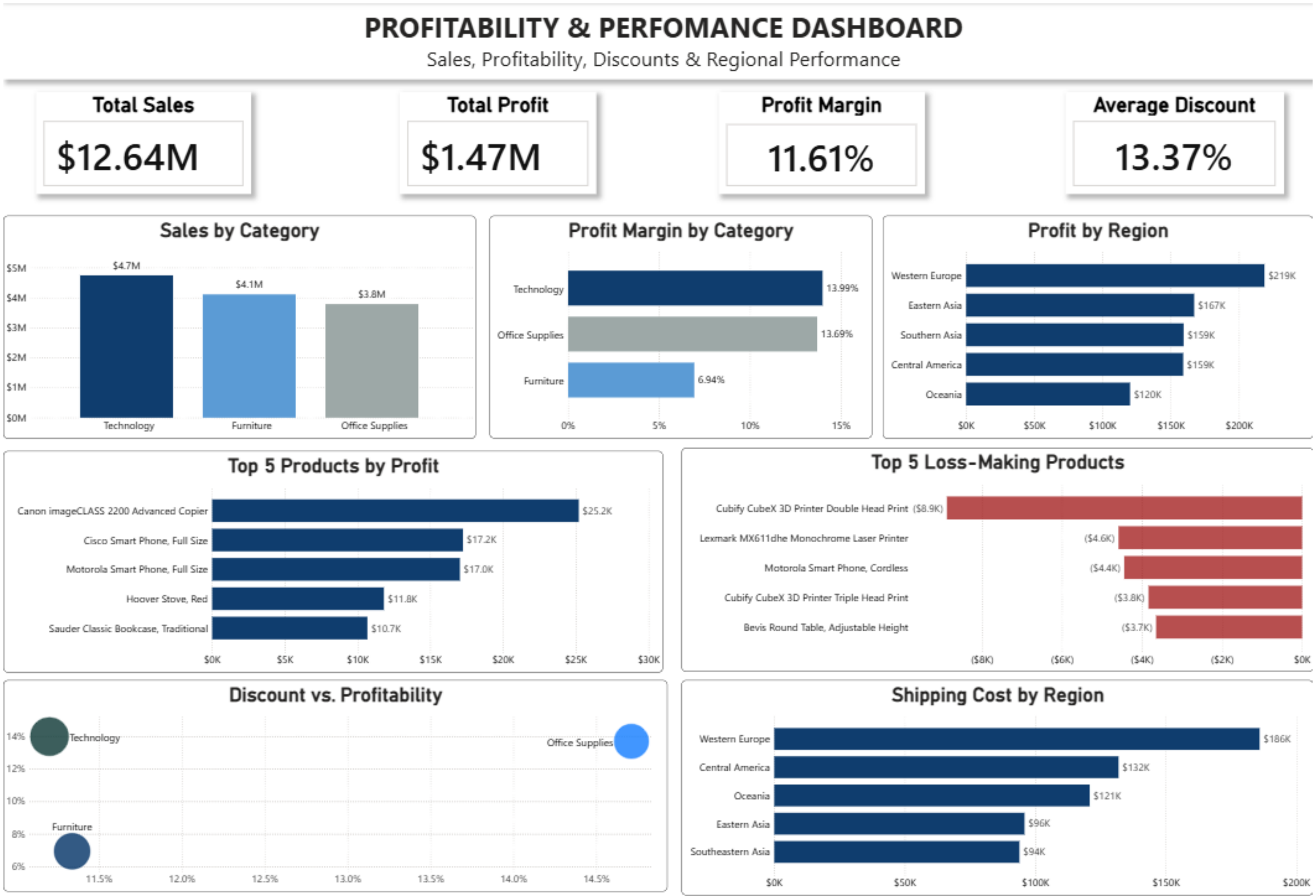

Figure 2: Global Superstore Profitability & Performance Dashboard

3.3 Product Findings

Finding	Result
Highest-Sales Product	Apple Smart Phone, Full Size — $86,936
Highest-Profit Product	Canon imageCLASS 2200 Advanced Copier — ~$25,200
Largest Identified Loss	Cubify CubeX 3D Printer Double Head Print — ~-$8,880
Highest Identified Margin	Avery 475 — 50.19% on $265 sales
Highest Quantity	Staples — 876 units

Product performance confirms that sales, profit and margin measure different aspects of value. Management should therefore evaluate products using multiple profitability indicators rather than revenue alone.

Note: The 50.19% margin for Avery 475 is based on only $265 in sales and therefore should not be interpreted as a major profitability driver.

4. Regional, Customer and Segment Performance

4.1 Regional Performance

Region	Sales	Profit / Indicator
Western Europe	$1.73M	~$0.22M
Central America	$1.22M	~$0.16M
Oceania	$1.10M	~$0.12M
Southeastern Asia	$0.88M	Leading group
Southern	$0.87M	Leading group

Western Europe is the strongest major region in the combined analysis. The SQL analysis recorded 2,993 orders, 22,262 products sold and $185,878 in shipping costs for Western Europe.

4.2 Canada as a High-Margin Benchmark

Canada recorded the highest regional profit margin in the SQL analysis at 26.61%, based on $66,932 in sales and $17,812 in profit. This is substantially above the overall company margin of 11.61%. Canada should therefore be treated as a high-margin benchmark rather than the strongest region by overall sales or profit. Management should investigate the pricing, product mix, discounting, shipping costs and customer characteristics that may be associated with Canada's stronger margin.

4.3 Customer Performance

Customer	Sales	Customer	Profit
Tom Ashbrook	$40,489	Tamara Chand	$8,676
Tamara Chand	$37,453	Raymond Buch	$8,454
Greg Tran	$35,552	Sanjit Chand	$8,203
Christopher Conant	$35,187	Hunter Lopez	$7,816
Sean Miller	$35,170	Bill Eplett	$7,408

Tom Ashbrook generated the highest sales, whereas Tamara Chand generated the highest profit. This shows why customer profitability should be assessed alongside revenue.

4.4 Customer Segments

Segment	Sales	Profit
Consumer	$6.51M	$749K
Corporate	$3.82M	$441K
Home Office	$2.31M	$277K

Consumer is the largest contributor to both sales and profit. Corporate and Home Office provide opportunities for targeted growth and account development.

5. Key Insights and Business Recommendations

5.1 Key Business Insights

- Technology drives profitability: it leads in category sales and profit and has a 13.99% margin.
- Furniture generates significant sales but has the weakest major-category margin at 6.94%.

- High sales do not always equal high profit at product, customer and monthly levels.
- Western Europe is the strongest major region by sales and profit.
- Canada's 26.61% margin provides a useful benchmark for investigating profitable practices.
- The Cubify CubeX 3D Printer Double Head Print has the largest identified product loss at approximately $8.9K.
- Discounting should be evaluated alongside profitability rather than sales alone. The Profitability & Performance Dashboard compares discount levels with profit margin while providing sales context, supporting further investigation into appropriate discount thresholds.
- Shipping costs are concentrated among a smaller group of regions, with the Top 5 Shipping Cost by Region visual highlighting priority areas for further investigation of logistics and fulfillment costs.
- Late-2015 demand indicates the need for stronger inventory and logistics planning.

5.2 Business Recommendations

1. Perform a detailed Furniture profitability review covering product costs, pricing, discounts, shipping and product mix.
2. Review major loss-making products and determine whether repricing, cost reduction, repositioning or discontinuation is appropriate.
3. Protect Technology's strong margin while pursuing profitable growth.
4. Evaluate discounts against incremental sales and profit to establish evidence-based discount thresholds.
5. Benchmark Canada's high-margin performance and identify transferable practices.
6. Optimize shipping by investigating high-cost regions and identifying the products, order patterns and fulfillment factors contributing to elevated shipping costs.
7. Prioritize customers using profit as well as sales when designing retention strategies.
8. Plan inventory, fulfillment and logistics ahead of seasonal demand.

5.3 Management KPI Baseline

KPI	Baseline
Total Sales	$12.64M
Total Profit	$1.47M
Overall Margin	11.61%
Average Discount	13.37%
Technology Margin	13.99%
Furniture Margin	6.94%
Canada Margin	26.61%
Largest Identified Product Loss	~-$8.9K

6. Limitations, Next Steps and Conclusion

6.1 Limitations

- The analysis covers historical data from 2012–2015 and therefore describes past performance.
- The dataset may not include supplier costs, marketing expenditure, inventory levels, competitor pricing or all operational drivers.

- The analysis identifies relationships and patterns but does not establish causation.
- Regional and product differences require deeper diagnostic analysis before major strategic decisions.
- Recommendations should be validated against current financial and operational information.

6.2 Recommended Next Steps

1. Analyze profit margin across discount bands to determine whether higher discounts are associated with weaker profitability and establish appropriate discount thresholds.
2. Measure shipping cost as a percentage of sales by product and region to distinguish high shipping volume from genuinely inefficient shipping costs.
3. Perform year-over-year sales, profit, margin and order growth analysis.
4. Investigate whether loss-making products are consistently unprofitable.
5. Conduct a detailed Furniture category diagnostic.
6. Investigate the drivers of Canada's high margin.
7. Develop customer profitability and retention analysis.
8. Analyze recurring seasonal patterns across the full analysis period.

6.3 Conclusion

The integrated SQL and Power BI analysis demonstrates that Global Superstore was profitable during 2012–2015, generating $12.64 million in sales and $1.47 million in profit at an overall margin of 11.61%. However, performance varies substantially across categories, products, regions, customers and time periods.

Technology is the strongest category, Furniture is a major profitability improvement opportunity, and Western Europe is the strongest major market. Canada's high regional margin provides a useful benchmark. Product-level analysis also identifies both major profit contributors and significant loss-making products.

The central management recommendation is to pursue profitable growth rather than sales growth alone. Better product margin management, discount discipline, regional benchmarking, logistics optimization, customer profitability analysis and seasonal planning can improve the amount of revenue converted into profit.

END OF REPORT